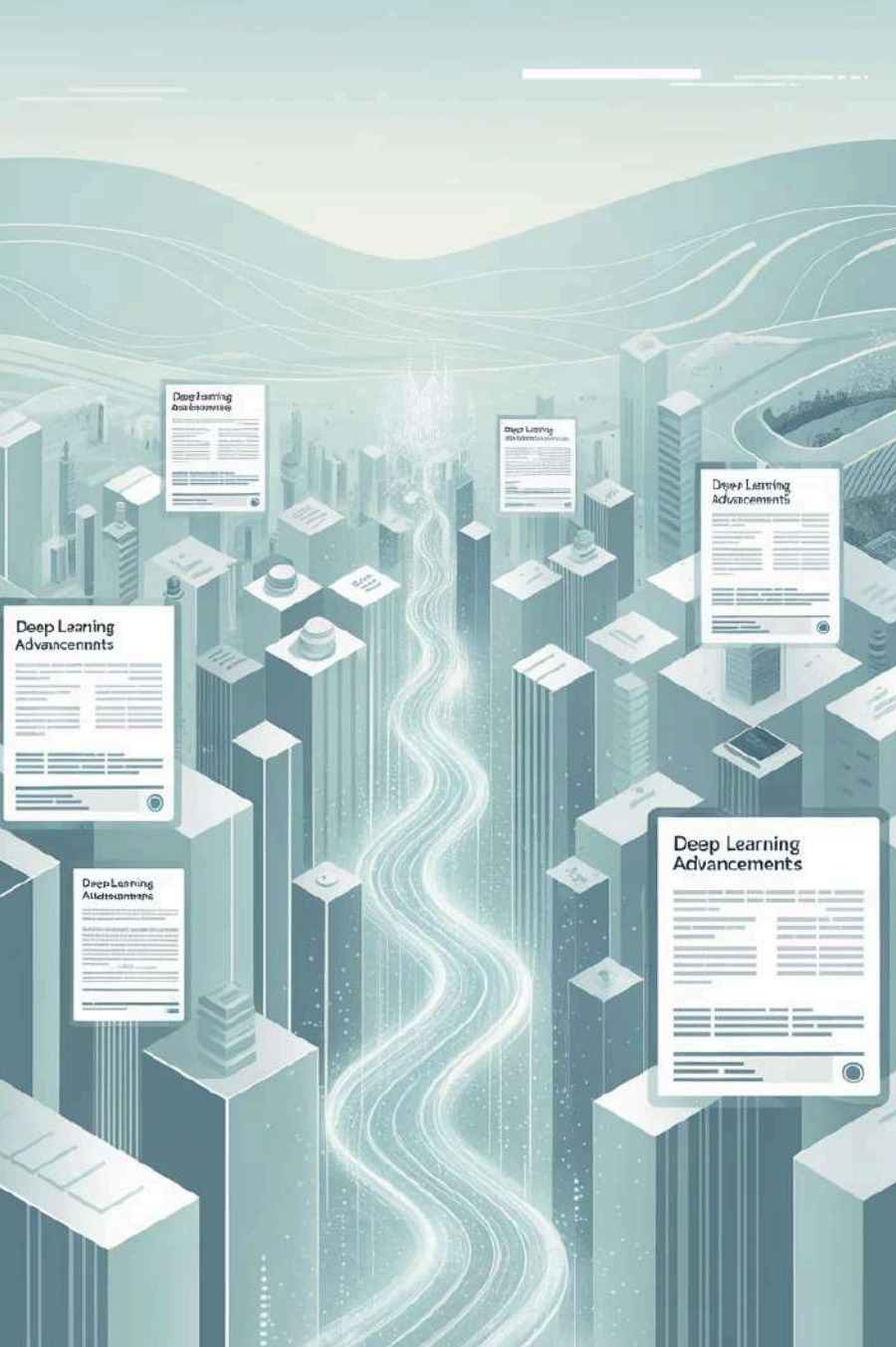




5.5 Years, 500+ Papers: A Fascinating Journey Through Deep Learning's Evolution

Meet Mike Erlihson(by some pseudo-random LLM)

A distinguished figure in the field of AI, known for his deep academic roots & significant contributions to AI content creation.

1

Academic Background

Holds a PhD in Math(stochastic processes), providing a strong theoretical foundation for his AI work.

2

Professional Leadership

Served as the Head of AI, leading innovative advancements in several companies.

3

Influential Creator

A prominent scientific content creator and AI influencer with over 64K+ followers, sharing insights and fostering community growth.

4

Research Expertise

Has conducted extensive paper reviews, analyzing over 500 DL papers(#DeepNightLearners project) - **the initial name**

5

Community Engagement

Hosts 2 popular podcasts, further engaging with the AI community, a co-author of the first ML/DL/AI book in Hebrew



Meet Mike Erlihson (by Mike)



Family

Married + 2 + some cats.



Hobby: Powerlifting & Workouts

Intensive workouts (calisthenics, military etc),
180-bench, 215-deadlift, 207.5-squat - 2007s.



Hobby: Architecture Enthusiast

Explore cities with cool architecture (Rome,
Prague, Barcelona, Vienna, etc).



Hobby: Scientific Reading

Read at least 1 paper + 1 blog daily, sometimes
recalling forgotten math.



Lifestyle: Intermittent Fasting

Eat within 6-7 hours window, anything desired.



Previous Hobby: Reading

Havr read tons of books (not scientific and not
math) until recently 🥲 - now only arxiv papers



Not a Big Fan Of...

Constructing stuff (not coding) by my own hands(paying handimen...)



DL Evolution: A Unique Perspective from my Reviews

Unique Dataset

528(and counting...) papers reviewed over 5.5 years represent a distinctive lens into DL's evolution. It's a story told through real-time research assessment.

We'll journey through 4 distinct "**Acts**" representing a fundamental shift in DL field's focus, substantiated by the papers I read during these years.

The 4 Acts of Our Journey

01

Act I: Foundations (Reviews 1-100)

Mastering GANs, Understanding Generalization, and Training Models w/o Tagged data

Perfecting fundamental tools, self-supervised learning & understanding how models learn & generalize, diffusion models start

03

Act III: LLM Gold Rush (Reviews 201-400)

A Massive Pivot to Reasoning, Alignment, and the Data That Fuels Them

Shift from building models to taming and understanding them.

02

Act II: Multimodal Shift (Reviews 101-200)

Diffusion Models Mature & the Hunt for a Transformer Successor Begins

Ambitious goals emerged as generative models expanded beyond single modalities.

04

Act IV: The Frontier (Reviews 401+)

From Models to Agents, Opening the Black Box, and Systematizing the Craft

Push toward active agents & some LLM understanding.

Act I: Foundations & Core Tools, Reviews 1-100



Mastering GANs

Focusing on stabilizing GANs, achieving high-resolution outputs, and tackling training instability and mode collapse challenges.



Understanding Generalization

Grappling with fundamental questions about model effectiveness, leading to counterintuitive discoveries challenging basic ML assumptions.



Self-Supervised Learning

SSL emerged, enabling models to learn effectively from vast amounts of unlabeled data, addressing data scarcity.

Generative Adversarial Networks (GANs): Recall



Core Concept

Two neural networks, a **Generator** and a **Discriminator**, compete in a zero-sum game to produce increasingly realistic data.



Dominant Model

Prior to diffusion models, GANs were the leading generative models, particularly for high-quality **image generation**.



Act I: Notable Papers That Crossed My Desk (Part 1)

On Generalization Peculiarities

Double Descent(Grokking): Challenged our assumptions, based on bias-variance tradeoff, that shook the community's scaling beliefs.

On Deep Neural Data Representation

Contrastive Learning: Self-supervision emerged as critical, with elegant theoretical frameworks explaining alignment & uniformity principles.



Act I: Notable Papers That Crossed My Desk (Part 2)

On Generative Models

Alias-Free GANs: The engineering focus remained on perfecting GAN architectures; addressing fundamental artifacts that had plagued generative image synthesis.

Adaptive Computation

PonderNet: : A neural architecture designed to adapt the amount of computation based on the complexity of the problem at hand.

The Turning Point

Diffusion Models Overtake GANs

A fundamental shift in generative modeling, signaling the start of a new era for creating realistic images.

A more stable, scalable, & powerful generative paradigm which redefined the field for years to come.



The Definitive Shift

"Diffusion Models Beat GANs on Image Synthesis"



- **Superior Sample Quality:** Unparalleled photorealism & crispness.
- **Better Mode Coverage:** Wider, more diverse output.
- **Enhanced Training Stability:** Robust and predictable process.
- **New Benchmark:** The leading architecture for high-fidelity synthesis.

Diffusion's Dominance

Witness the new era of image synthesis:
unparalleled realism and boundless creative
potential.





Decipher 4 first reviews permutation: chronologically

- Are Deep Neural Architectures Losing Information? Invertibility Is Indispensable?
- Deep Double Descent: Where Bigger Models and More Data Hurts
- Curriculum by Smoothing
- Contrastive Representation Distillation

Think carefully before you answer!



$$\pi = \{3, 4, 1, 2\}$$

1. Curriculum by Smoothing
2. Contrastive Representation Distillation
3. Are Deep Neural Architectures Losing Information? Invertibility Is Indispensable?
4. Deep Double Descent: Where Bigger Models and More Data Hurts



Act II: Diffusion Everywhere, Multimodality & Transformer Successor Race, Reviews 101-200

Diffusion Matures

Diffusion models became controllable and practical, with text-guided generation leading to photorealistic results.

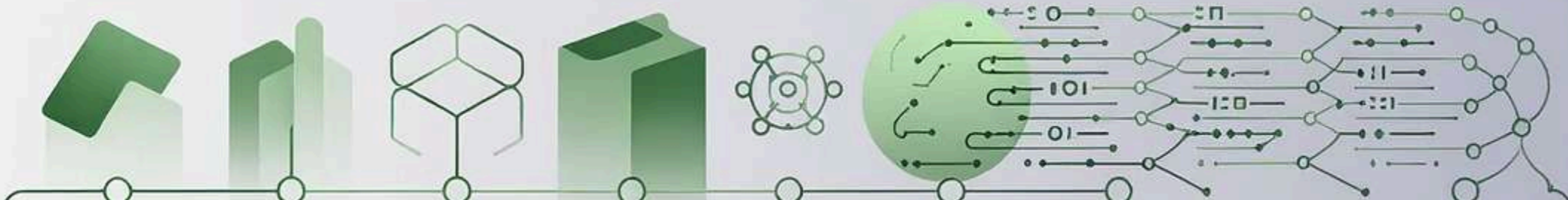
Multimodality is King

Unified models processing images, text, and audio simultaneously drove cross-modal understanding and generation.

The Great Successor Race

A race for Transformer successors led to novel, more efficient sequence modeling architectures.

EVOLUTION



Act II: Breakthrough Papers I Evaluated

1

Text-to-Image-Video Revolution

GLIDE/MakeAVideo: Diffusion's maturation into controllable, photorealistic image generation guided by natural language descriptions(felt almost magical)

2

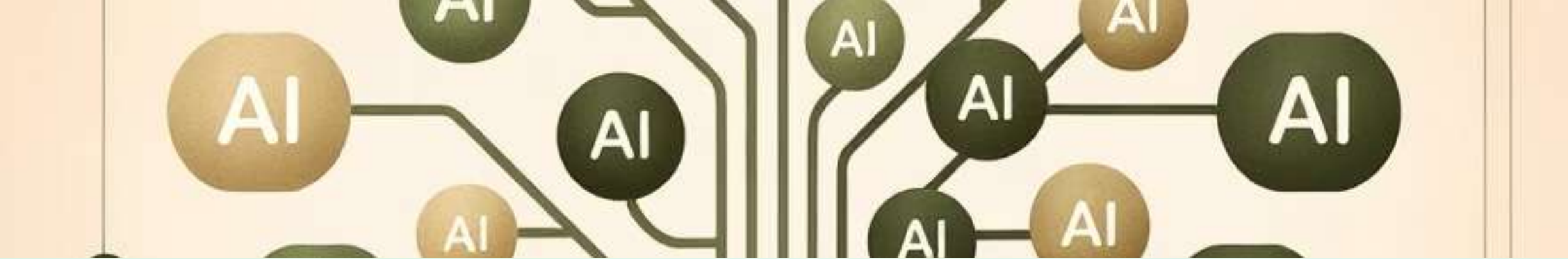
Multimodal Ambitions

SeamlessM4T: Exemplified the field's goal to unify human communication across languages and modalities in a single, elegant framework for translation & understanding.

3

Early Reasoning Signals

Graph-of-Thought: The first signs of the next wave appeared with complex reasoning approaches that went beyond simple CoT to capture complex logical relationships.



The Cambrian Explosion of Architectures: Part 1

A battlefield of architectural innovation: Next Transformer "Killers"

Hyena Hierarchy

Attempting to scale convolutional approaches to language modeling, promising better efficiency for longer sequences through hierarchical convolutions.

Retentive Networks, RWKV

A bold claim to be the direct successor to Transformers, balancing the training efficiency of parallelism with the inference efficiency of recurrence.

The Cambrian Explosion of Architectures: Part 2

1

Mamba: The Breakthrough

Felt the first truly viable, widely-adopted alternative achieving linear-time sequence modeling with selective state spaces and genuine practical impact.

2

Kolmogorov-Arnold Networks (KAN)

Replaced traditional non-linearities with learnable activation functions, representing a fundamental rethinking of neural net architecture.

Act III: The LLM Gold Rush, Reviews 201-400

From Building Models to Taming Them

Paradigm Shift: Raw model capability was no longer enough: make models reliable, safe & useful in real-world applications.



Reasoning & Alignment

Teaching models to think clearly & follow human intentions became the central challenge, requiring sophisticated training techniques & evaluation methods.



Data Quality Awakening

Better data, not just more data: key to capability improvements; quality over quantity became the mantra.



Critical Evaluation

Healthy skepticism emerged around scaling laws & emergent abilities, demanding more rigorous evaluation & honest assessment of model limitations.

Teaching Models to Think

Enabling models to engage in genuine reasoning rather than pattern matching. How models could bootstrap their own reasoning capabilities?

STaR: Self-Taught Reasoner

Models could generate their own training data to improve reasoning, a recursive approach to capability enhancement that felt almost like artificial self-improvement.



Quiet-STaR Evolution

Evolved into more sophisticated internal monologue systems, where models learned to "think before speaking" through continuous self-generated reasoning traces.

Moving from external reasoning techniques to models developing their own internal reasoning processes towards genuine artificial cognition.



Alignment & Data Quality Revelations

The Alignment Debate

- 1 "Is DPO Superior to PPO for LLM Alignment?" - a paradigm shift toward simpler, offline alignment techniques that eliminated complex RL pipelines

The Data Warning: A wake-up call

- 2 "AI models collapse when trained on recursively generated data", highlighted the hidden dangers of model collapse & emphasized the crucial need for data hygiene in an era of synthetic training data.

These revelations fundamentally changed how the community approached training data curation and model alignment strategies.



The Critical Lens Emerges

Developed the confidence to critically examine its own claims & identify fundamental limitations rather simply pursuing capability increases.

On Scaling Laws

"Are Emergent Abilities of LLMs a Mirage?": essential rigor to LLM evaluation, questioning whether claimed breakthroughs were genuine phase transitions or measurement artifacts.

On Inherent Limitations

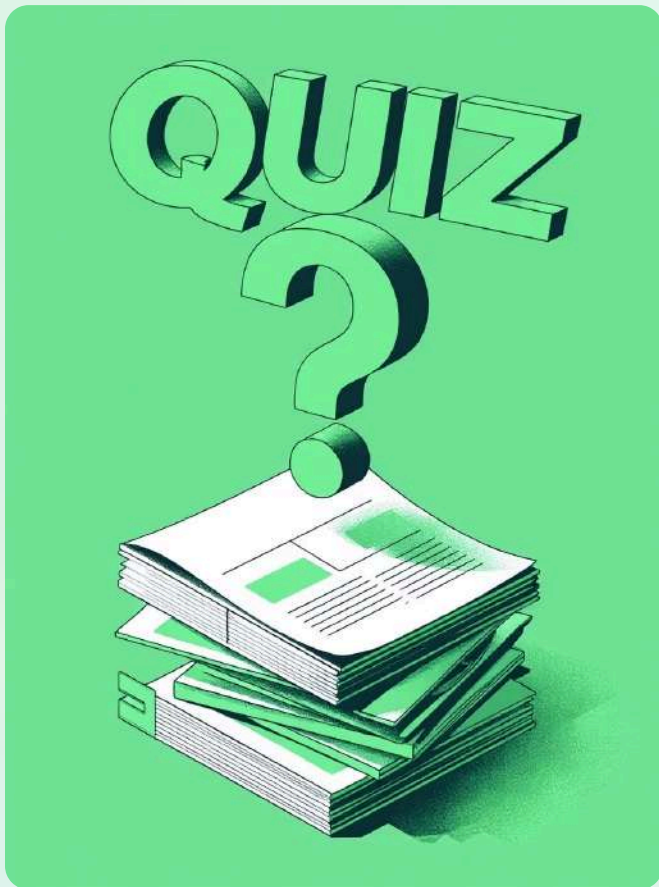
"LLMs Will Always Hallucinate, We Need to Live With This": a pragmatic shift toward acknowledging & working with fundamental model limitations rather than trying to eliminate them entirely.



GSM-Symbolic: Understanding Math's LLM Limits

Linguistically fluent LLMs, struggle with true math reasoning & symbolic logic, often relying on statistical patterns over genuine deduction.

Highlights a fundamental limitation & suggests integrating symbolic methods to build more intelligent systems.



Decypher the wordcloud permutation

Word frequency in my reviews:

המחברים

שפה

דאטה

מודל

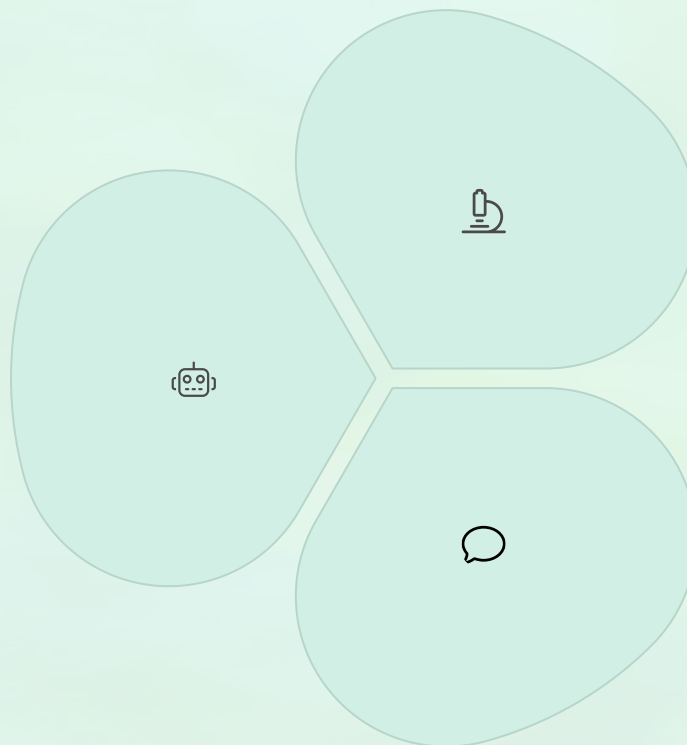
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Act IV: The Current Frontier, Reviews 401+

Pushing Toward What's Next

Represents a fundamental evolution in ambition, moving beyond static content generation toward dynamic, interactive systems that can perceive, reason, and act in complex environments.

Active Agents
Models that complete tasks through actions, not just generate content

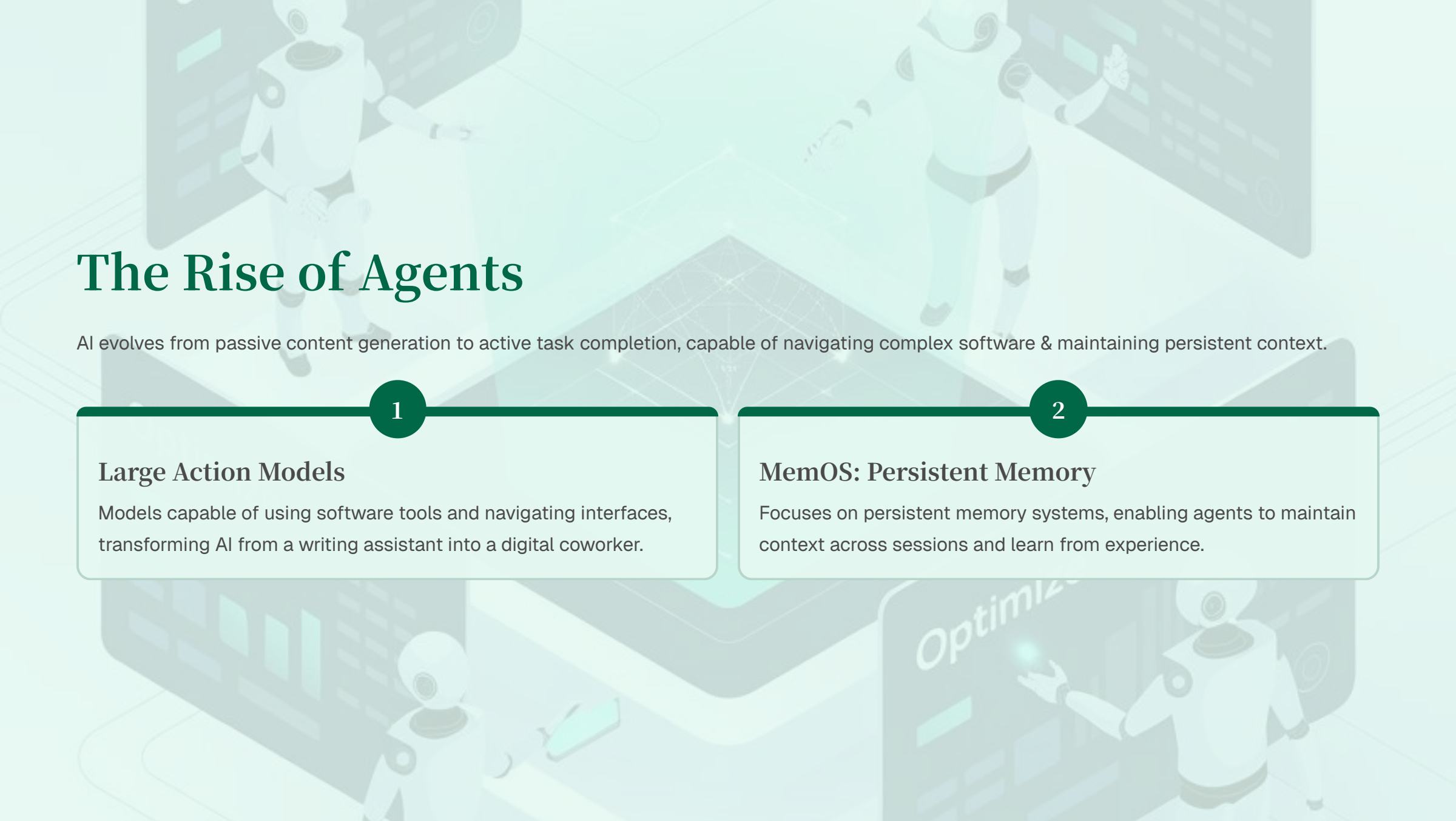


Mechanistic Understanding

Opening the black box to understand how models actually work internally

Systematic Craft

Maturing from art to science with principled approaches to optimization

The background features a stylized, isometric illustration of three humanoid AI agents in a digital space. One agent is in the top left, another in the top right, and a third in the bottom right. They are surrounded by floating data visualizations, including bar charts, line graphs, and a large circular radar chart in the center. The overall color palette is a mix of light blues, greens, and greys, giving it a futuristic, tech-oriented feel.

The Rise of Agents

AI evolves from passive content generation to active task completion, capable of navigating complex software & maintaining persistent context.

1

Large Action Models

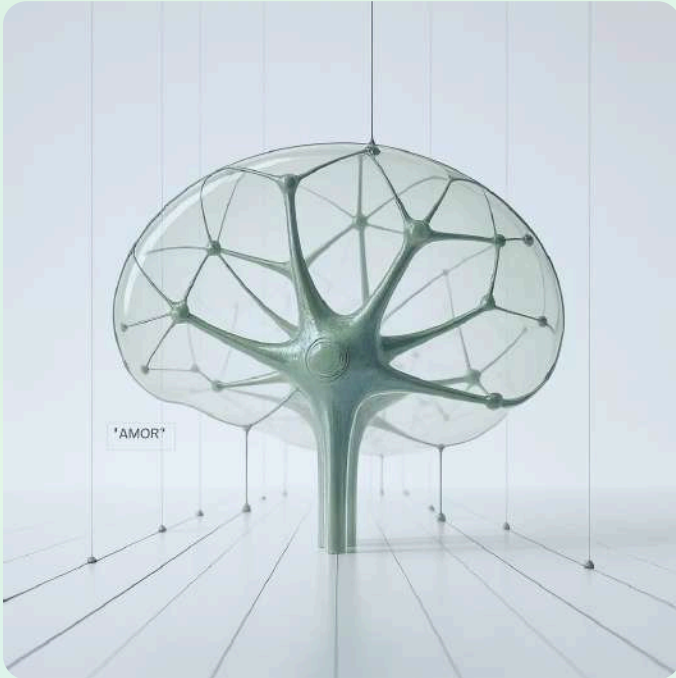
Models capable of using software tools and navigating interfaces, transforming AI from a writing assistant into a digital coworker.

2

MemOS: Persistent Memory

Focuses on persistent memory systems, enabling agents to maintain context across sessions and learn from experience.

Opening the Black Box of LLMs



Mechanistic interpretability is a major research thrust, focusing on understanding internal model computations.

The Geometry of Concepts: The Anthropic "Circuits"

Sparse Autoencoder Feature Structure (e.g., Anthropic "Circuits") exemplifies this by identifying human-understandable concepts within model activations.

Defining Grand Challenges

The field has matured, systematically articulating its fundamental problems, as seen in "**Open Problems in Mechanistic Interpretability**"—a roadmap for AI cognition.



Fundamental Innovation Continues

Pushing architectural boundaries & seeking more fundamental theoretical understanding of DL systems.



New Theoretical Foundations

Rethinking Transformers Through Physics: An attempt to ground neural architectures in energy-based models, seeking a new theoretical foundations beyond attention mechanisms



Optimizing the Training Engine

Fantastic Pretraining Optimizers: Refining the fundamental engine of DL itself, with recent papers finding significant improvements in training dynamics and convergence properties.



4 Overarching AI Future Trends & A Single Small Trend



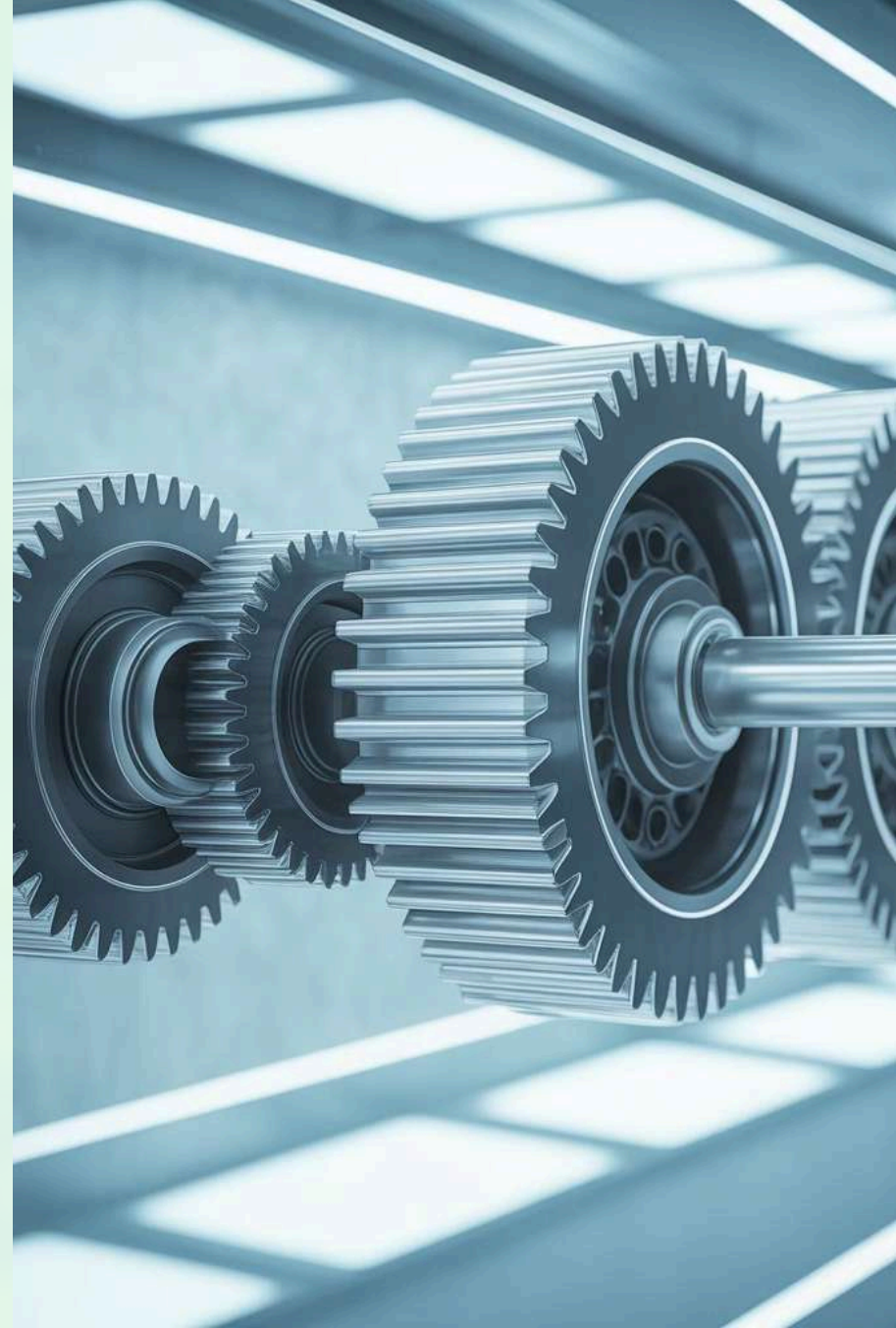
AMOR

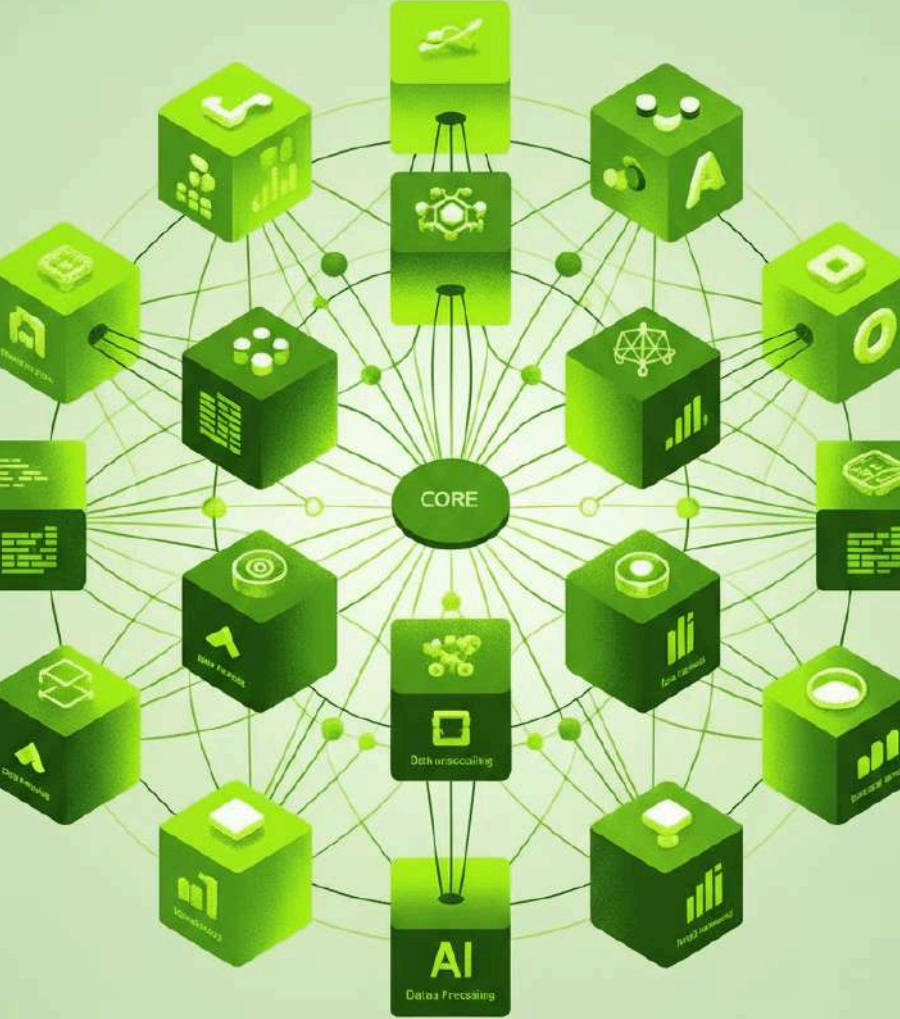
Trend 1: Industrialization of Alignment

A niche RLHF research has transformed into standardized, engineering-heavy practices central to all major AI labs. Alignment moved from academic curiosity to production necessity.

Trend 2: Efficiency Over Raw Scale

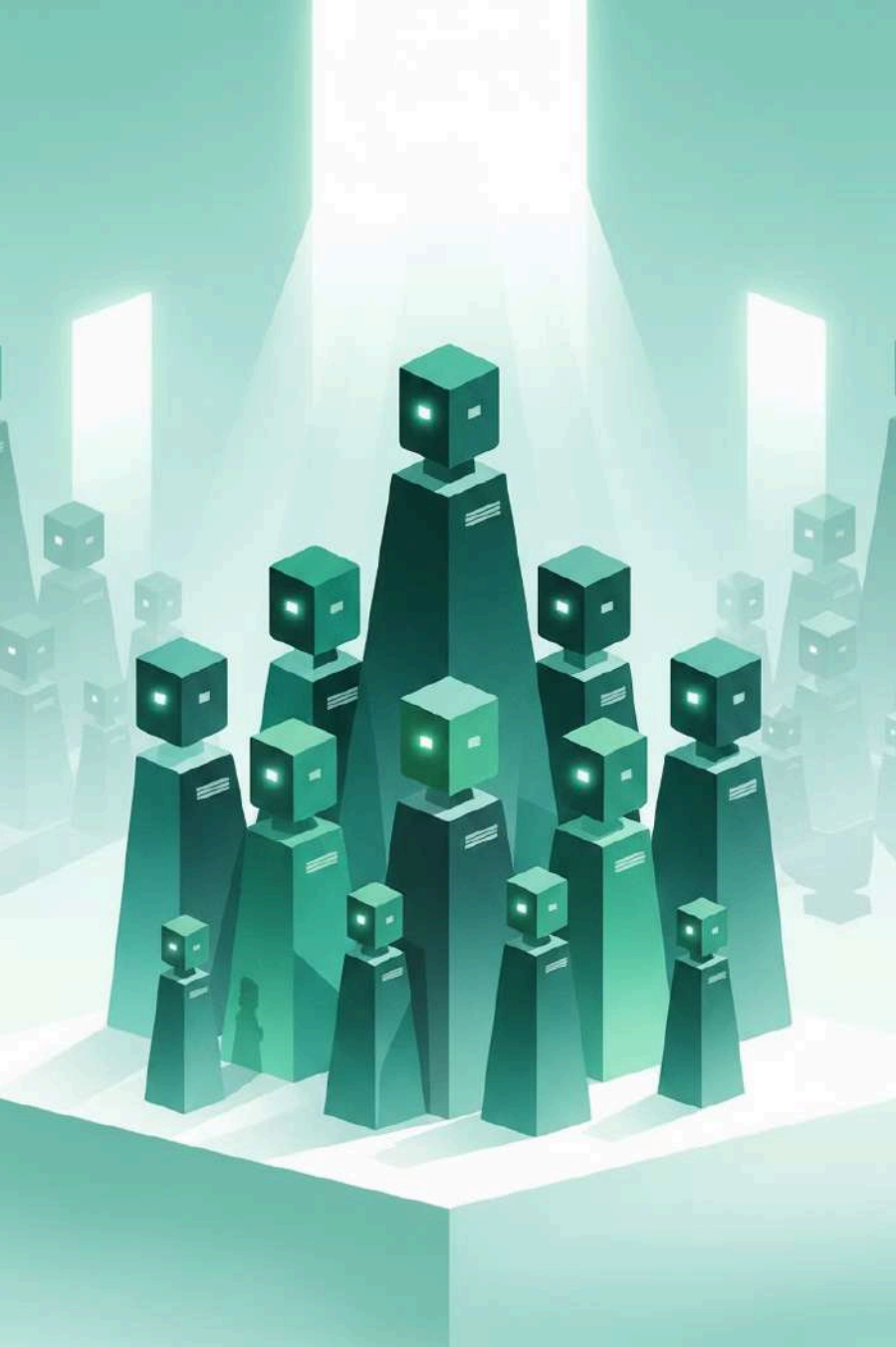
The brute-force scaling approach is being replaced by nuanced focus on high-quality data curation & architecturally efficient designs. Smarter training trumps bigger training.





Trend 3: Unbundling the Monolith

Transition from single giant models toward sophisticated ecosystems of specialized models, agents, memory systems & tools working in coordination.



Trend 4: Small Models Within Agents Per Application

A rise in smaller, highly focused models embedded within intelligent agents, tailored to excel at a specific task/ application.

Greater efficiency, accuracy, & customizability, moving away from a 1size-fits-all solution towards a distributed network of AI components optimized for their particular domain.

A Small Trend 1: Fast Text (Image, Video) Generation Models

Language diffusion models, speculative decoding, quantization & other parallel generation techniques are enabling real-time applications & highly interactive AI experiences.

SynapseA



Real-time.
Reimagined

My Prediction (Based on 528 Reviews)

The next major capability leap won't come from scaling alone.

Breakthroughs in **long-context reasoning systems** combined with efficient RAG techniques & smart memory management

Development of "**world models**" that can simulate and plan, themes increasingly prominent in my recent review queue.



Next Goal: The 1024-Paper Reviews Milestone

496 left and counting down, meet you at 777....





Thank You

Questions?